Hydroponics & Food Production: System Design Description

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1. System overview

Hydroponics and food production grow fresh produce for the resident crew on stacked grow tiers under controlled lighting, and prepare meals in the central kitchens.

The system is led by Agriculture & Food Systems. It is delivered through 6 work packages with a combined budget of 85,718.02 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
AG-HGT	Hydroponic grow tier	SHV-AG-SPEC-0101	SHV-AG-DWG-0201	SHV-AG-BOM-0301
AG-NSD	Nutrient solution dosing unit	SHV-AG-SPEC-0102	SHV-AG-DWG-0202	SHV-AG-BOM-0302
AG-LGT	Grow lighting array	SHV-AG-SPEC-0103	SHV-AG-DWG-0203	SHV-AG-BOM-0303
AG-FPK	Food processing kitchen	SHV-AG-SPEC-0104	SHV-AG-DWG-0204	SHV-AG-BOM-0304

3. Architecture

3.1 Hydroponic grow tier

The hydroponic grow tier is specified in SHV-AG-SPEC-0101. Its governing parameter is the growing area, set at 6,934 m². The responsible engineer is Carys Renner.

3.2 Nutrient solution dosing unit

The nutrient solution dosing unit is specified in SHV-AG-SPEC-0102. Its governing parameter is the dosing accuracy, set at 2.4 %. The responsible engineer is Carys Renner.

3.3 Grow lighting array

The grow lighting array is specified in SHV-AG-SPEC-0103. Its governing parameter is the photon flux, set at 947 µmol/m²/s. The responsible engineer is Joren Mathers.

3.4 Food processing kitchen

The food processing kitchen is specified in SHV-AG-SPEC-0104. Its governing parameter is the meals per day, set at 68,391 ea. The responsible engineer is Elara Ingram.

4. Interfaces

Document	Title
SHV-ICD-PD-AG	Interface Control Document: Power Distribution / Hydroponics & Food Production
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-AG-01	Hydroponics & Food Production: Design & analysis	Elara Ingram	IN_PROGRESS	90.5%
WP-SHV-AG-02	Hydroponics & Food Production: Fabricate & install hydroponic grow tier	Elara Ingram	IN_PROGRESS	87.8%
WP-SHV-AG-03	Hydroponics & Food Production: Fabricate & install nutrient solution dosing unit	Treva Zander	IN_PROGRESS	92.0%
WP-SHV-AG-04	Hydroponics & Food Production: Fabricate & install grow lighting array	Tamsin Ingram	IN_PROGRESS	91.3%
WP-SHV-AG-05	Hydroponics & Food Production: Fabricate & install food processing kitchen	Faren Hesk	IN_PROGRESS	83.4%
WP-SHV-AG-06	Hydroponics & Food Production: Integration & test	Alira Corrin	IN_PROGRESS	75.5%

6. Risks

- Supply of hydroponic grow tier assemblies depends on a single supplier.
- Supply of nutrient solution dosing unit assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.